

DILEEP SIVARAMAN

Biomedical Engineer | Medical Robotics

Present Affiliation:

Foreign Expert,

Center for Biomedical and Robotics Technology
(BART LAB),
Faculty of Engineering,
Mahidol University,
Salaya, Thailand 73170

Email ID: dileepsivaraman89@gmail.com

: dileep.siv@mahidol.ac.th

Phone: +66843410548

HOME:

Dileep Bhavanam
Kadampanad South PO
Pathanamthitta (DST)
Kerala,
Pin- 691553
India



Google Scholar: https://scholar.google.com/citations?user=R_I4Ut0AAAAJ&hl=en&oi=ao

Website: <https://dileepsivaraman007.github.io/dileepsivaraman/>

RESEARCH PROFILE

Biomedical engineering researcher working at the intersection of medical robotics, bio-inspired mechanisms, physical-system modeling, and probabilistic state estimation. Research emphasizes minimally invasive robotic systems, sensorless and embedded force estimation, smart-material behavior, system identification, and the use of system dynamics for sensing, estimation, and control.

Core areas: Medical robotics; bio-inspired robotic mechanisms; embedded sensing and force estimation; smart materials; robotic-system modeling

Methods: Kalman filtering; adaptive polynomial filtering; sensor fusion; nonlinear state estimation; time-series modeling; system identification; adaptive filtering; uncertainty quantification

ACADEMIC APPOINTMENTS AND RESEARCH EXPERIENCE

2025-Present	Foreign Expert BART LAB, Faculty of Engineering, Mahidol University Conduct research on data-driven estimation and modeling of robotic and bio-inspired systems.
2020-2025	Ph.D. Researcher Mahidol University - Developed adaptive state-estimation frameworks for bio-inspired needle systems in minimally invasive surgical robotics, with application to breast biopsy. - Contributed to multi-robot collaboration research, developing coordinated planning and decision-making strategies for heterogeneous robotic systems in disaster-response environments.
2017-2019	Lecturer Department of Physics, Shree Vidyadhiraja College of Arts & Science, Kerala University, India
2017-2019	National Service Scheme Programme Officer Shree Vidyadhiraja College of Arts & Science, Kerala, India
2013-2017	Lecturer Al-Azhar Group of Institutions, Department of Technical Education, Kerala, India

EDUCATION

17 February 2025	Ph.D., Biomedical Engineering (Medical Robotics) Mahidol University, Thailand
2012	Integrated M.Sc., Physics and Mathematics Amrita Vishwa Vidyapeetham University, India

SELECTED RESEARCH PROJECTS AND CONTRIBUTIONS

2025-Present	Nonlinear Hysteresis-Aware State Estimation for SMA-Based Minimally Sensed Surgical Systems - Developing an SMA-based embedded force-sensing mechanism for minimally invasive bio-inspired needle systems. - Investigating nonlinear electromechanical and hysteretic behavior of SMA elements under constrained insertion dynamics. - Formulating a physics-constrained state-estimation framework for real-time internal-force reconstruction under limited sensing.
2021-2025	Bio-Inspired Needle for Breast Biopsy Using a Novel State Estimation/Prediction Algorithm Integrated robotic-system design with clinical requirements for biopsy procedures is supervised by Dr. Songpol Ongwattanakul (BART LAB), Dr. Jackrit Suthakorn, (BART LAB) and Prof. Cholatip Wiratkapun, Department of Radiology, Ramathibodi Hospital.
2023-Present	Advanced Research in Adaptive Polynomial Kalman Filtering and Optimization - Collaborative project with Prof. Asoke Nandi, Brunel University London, focused on nonlinear state estimation in time-varying systems. - Developed and analyzed model-order selection methods, mu-term augmentation, and hybrid autoregressive-polynomial structures.
2026-Present	Task-Oriented Link-Length Optimization for Mobile Manipulators in Disaster Response Conducting kinematic design and workspace optimization for multi-task disaster-response scenarios, including RoboCup Rescue-style manipulation tasks.
2022-2023	e-ASIA Joint Research Program: Disaster Risk Reduction Management Contributed to an international collaboration with Kazan Federal University and Kyoto University on unmanned systems, terrain modeling, sensor integration, and AI-driven robotic strategies for harsh environments.
2022-2023	Development of Unmanned Aerial Vehicles for Disaster Risk Reduction Management Developed a bio-inspired path-planning strategy for heterogeneous robotic systems operating in uncertain terrain.
2022	Stair-Climbing Wheelchair with Navigation and Mapping Assisted with mechanical-component assembly, integration, and testing.
2020	Mask Material Filtration Efficiency Testing Faculty of Medicine, Siriraj Hospital, Mahidol University Conducted testing in coordination with medical professionals and hospital-grade requirements.

SCHOLARLY PUBLICATIONS

Peer-Reviewed Journal Articles

- Sivaraman, D., et al. "Dynamic order selection analysis in adaptive polynomial Kalman filtering: implementation and integration of sensor data and hybrid image processing for bio-inspired needle systems." **Systems Science & Control Engineering** 13(1) (2025): 2546839.
- Kunapinun, A., et al. "Toward AI-Powered Neurovascular Intervention: From Imaging to XR-Robotic Convergence." **Stroke**.
- Sivaraman, D., Pillai, B. M., Ongwattanakul, S., & Suthakorn, J. "Adaptive Polynomial Predictive Filter for Sensor Data Estimation and Prediction in Interference Environment." **IET Control Theory & Applications** (2024).
- Sivaraman, D., Ongwattanakul, S., Moonjaita, C., Suthakorn, J., & Pillai, B. M. "A pack hunting strategy for heterogeneous robots in rescue operations." **Bioinspiration & Biomimetics** (2024).
- Sivaraman, D., Ongwattanakul, S., Pillai, B. M., & Suthakorn, J. "Adaptive polynomial Kalman filter for nonlinear state estimation in modified AR time series with fixed coefficients." **IET Control Theory & Applications** (2024).
- Pillai, B. M., Suthakorn, J., Sivaraman, D., et al. "A heterogeneous robots collaboration for safety, security, and rescue robotics: e-ASIA joint research program for disaster risk and reduction management." **Advanced Robotics** 38(3) (2024): 129-151.
- Pillai, B. M., Owatchaiyapong, P., Treratanakulchai, S., Sivaraman, D., et al. "Lower Limb Exoskeleton with Energy-Storing Mechanism for Spinal Cord Injury Rehabilitation." **IEEE Access** 11 (2023): 133850-133866.

8. Sivaraman, D., Ongwattanakul, S., Suthakorn, J., & Pillai, B. M. "Nonlinear Dynamic States' Estimation and Prediction Using Polynomial Predictive Modeling." **IEEE Canadian Journal of Electrical and Computer Engineering** 46(3) (2023): 185-195.

International Conference Proceedings

1. Nakdhamabhorn, S., & Sivaraman, D. "A Parallel Ant-Colony Pheromone-Inspired Path Planner for Assistive and Hospital Service Robot Navigation." Accepted for oral presentation, Biomedical Engineering International Conference (**BMEiCON 2026**), Fukuoka, Japan, 10-13 October 2026. Paper 1571321925. Forthcoming.
2. Moonjaita, C., et al. "Ground Reaction Force Analysis of Robot-Assisted Walking Using Inertial Sensors." 18th International Convention on Rehabilitation Engineering and Assistive Technology (**i-CREATE**), IEEE (2025).
3. Sivaraman, D., Wiratkapun, C., Pillai, B. M., Suthakorn, J., & Ongwattanakul, S. "Shape Memory Alloy Spring-Based Force Sensing System for Dual-Sheath Surgical Needle." IEEE International Conference on Robotics and Biomimetics (**IEEE ROBIO**), Chengdu, China (2025).
4. Sivaraman, D., Wiratkapun, C., Pillai, B. M., Suthakorn, J., Methachan, B., Nakdhamabhorn, S., & Ongwattanakul, S. "Hybrid image-based motion tracking and adaptive polynomial Kalman filter for a bio-inspired dual-sheath needle system." **IEEE ROBIO**, Bangkok, Thailand (2024).
5. Pillai, B. M., Siripala, N., Sai-aroon, K., Sivaraman, D., Vinjamuri, R., Chumnanvej, S., & Suthakorn, J. "Enhancing post-stroke upper limb rehabilitation through haptic feedback in virtual reality-based gaming." **IEEE ROBIO**, Bangkok, Thailand (2024).
6. Sivaraman, D., et al. "Adaptive Polynomial Predictive Filter: Solving Inconsistent and Interrupted Sensor Data Challenges." **IEEE ROBIO**, Ko Samui, Thailand (2023).
7. Sivaraman, D., et al. "Energy Optimized Path Planning and Decision Making for Multiple Robots in Rescue Operations." **IEEE IECON** 2022, pp. 1-6, Brussels, Belgium (2022).
8. Sivaraman, D., et al. "Sensorless Based Gravity Torque Estimation and Friction Compensation for Surgical Robotic System." **IEEE ICELIE**, pp. 1-6, Brussels, Belgium (2022).
9. Sivaraman, D., et al. "Energy Optimized Path Planning and Decision Making for Multiple Robots in Rough Terrain." **AROB 27th / ISBC 7th / SWARM** 5th, online, 25-27 January 2022.

Peer-Reviewed Local Proceedings

1. Sivaraman, D., et al. "Curvature Estimation of Steerable Flexible Probe for Ductal Carcinoma in Situ Detection: A Simulation Study." 18th Asian Conference on Computer Aided Surgery and 5th International Conference on Business, Informatics, and Management (August 2022).

Book Chapter

2. Sivaraman, D., Pillai, B. M., Suthakorn, J., & Ongwattanakul, S. (2024). "Human-Robot Interaction in Biopsy Procedures: A Biomimetic Dual-Sheath Needle Design Inspired by Insect Ovipositor Mechanics." In *Discovering the Frontiers of Human-Robot Interaction: Insights and Innovations in Collaboration, Communication, and Control* (pp. 401-417). Springer Nature Switzerland.

Manuscripts Under Review or Submitted

3. Sivaraman, D., Pillai, B. M., Suthakorn, J., Nandi, A. K., & Ongwattanakul, S. "Adaptive Polynomial Kalman Filtering and Model Order Selection: Comparative Analysis with Polynomial and Autoregressive Representations." **Digital Signal Processing: A Review Journal**. Under review.
4. Sivaraman, D., Pillai, B. M., & Suthakorn, J. "Error-Dependent Bandwidth Shaping for DC Motor Control via Analytical Pole Placement." **Mechatronics**. Submitted.
5. Sivaraman, D., Pillai, B. M., & Suthakorn, J. "Direction-Dependent Friction Identification and Compensation in Brushed DC Servo Motors Using Closed-Loop Disturbance Observer Data." **IEEE Transactions on Industrial Electronics**. Submitted.

FELLOWSHIPS, GRANTS, AND ACADEMIC SUPPORT

- Travel Grant, Faculty of Engineering, Mahidol University, for presentation at IEEE ROBIO 2025, Chengdu, China, 3-7 December 2025.
- Research support through the Reinventing University System and Mahidol University (IO 864102063000) for Integrating Navigation and Robotics for Breast Biopsy Using Imaging and Models (Pre-Clinic and Clinic Evaluation Phase for Product Commercialization), 2023-2024.
- Mahidol Postgraduate Scholarships for Graduate International Students, 2020-2023.

TEACHING AND MENTORSHIP

Foreign Expert Teaching | Department of Biomedical Engineering, Mahidol University

EGBI481 Introduction to Medical Robotics	Undergraduate BME Aug-Dec 2025	Shared with Prof. Jackrit Suthakorn
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EGBE680 Advanced Medical Robotics	Graduate BME Jan-May 2026	Shared with Prof. Jackrit Suthakorn
EGBI481 Introduction to Medical Robotics	Undergraduate BME Aug-Dec 2026	Shared with Prof. Jackrit Suthakorn

Teaching Collaboration | Asian Institute of Technology, Thailand

Self-Driving Vehicles in Smart City	Graduate 2026	Shared with Dr. Branesh Madhavan Pillai
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Doctoral Teaching Assistantship | Department of Biomedical Engineering, Mahidol University

Computational Methods for Biomedical Engineering (Python)	Undergraduate BME 2023-2024	Shared with Dr. Songpol Ongwattanakul
Healthcare Automation	Graduate BME 2024	Shared with Dr. Songpol Ongwattanakul

ACADEMIC LEADERSHIP AND PROFESSIONAL SERVICE

- Section Chair, IEEE International Conference on Robotics and Biomimetics (ROBIO 2025).
- Academic Committee Member, 18th International Convention on Rehabilitation Engineering and Assistive Technology (i-CREATe 2025), November 2025.
- IOP Peer Review Excellence Certificate, Institute of Physics Publishing, 2025.
- BART LAB Robotics Team member, RoboCup 2023, Bordeaux, France, 4-10 July 2023.
- Technical Volunteer, RoboCup 2022, Bangkok.

INVITED SPEAKING

- Keynote Speaker, 1st International Conference on AI Innovations in Medical Engineering and Robotics Technology (AI-MERT 2025), Asian Institute of Technology, Thailand; event scheduled for 18-19 December 2025.

TECHNICAL SKILLS

Programming and robotics: MATLAB, Python, ROS
Mechanical design: SolidWorks, Onshape
Research communication: LaTeX, academic publications, technical documentation, research reports
Web and version control: Git, GitHub Pages, Google Sites, HTML/CSS

THESIS AND EARLIER RESEARCH

Integrated M.Sc. thesis: Photometric Studies of Galactic Open Star Cluster Berkeley 7. Supervisor: Dr. Sneha Lata, Aryabhata Research Institute of Observational Sciences, Department of Science and Technology, Government of India, Manora Peak, Nainital, India.
University project: Dynamics of Nonlinear Maps. Supervisor: Dr. V. M. Nandakumaran, Head, Department of Physics, Amrita Vishwa Vidyapeetham University, Amritapuri Campus, Kerala, India.
Scholarly training: Workshop on Recent Trends in Physics (2011), organized by the Indian Academy of Sciences, National Academy of Sciences, and Indian National Science Academy.

COMMUNITY ENGAGEMENT

- Flood-relief volunteer, Plan International (India Chapter), Kerala, 2018.

PROFESSIONAL REFERENCES

Dr. Songpol Ongwattanakul

Executive Director, Center for Biomedical and Robotics Technology (BART LAB), Department of Biomedical Engineering, Faculty of Engineering, Mahidol University, Salaya, Thailand 73170
songpol.ong@mahidol.ac.th | <https://bartlab.org/>

Dr. Branesh Madhavan Pillai

Assistant Professor, Department of Industrial Systems Engineering, School of Engineering and Technology, Asian Institute of Technology, Pathum Thani 12120, Thailand
branesh@ait.asia | <https://www.ait.ac.th/>

Prof. Cholatip Wiratkapun, M.D.

Professor of Radiology and Head of Department, Faculty of Medicine Ramathibodi Hospital, Mahidol University, Bangkok 10400, Thailand
cholatip.wir@mahidol.ac.th | cholatipp@gmail.com

Dr. Ramana Vinjamuri

Associate Professor of Computer Science and Electrical Engineering, University of Maryland, Baltimore County, 1000 Hilltop Circle, Baltimore, MD 21250, USA
rvinjam1@umbc.edu